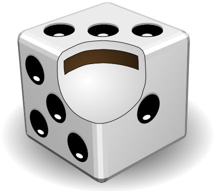
Simple Games

- Games: a mathematical formalism for rational interaction
- What is the most popular solution concept? (Nash Equilibrium)

Types of Uncertainty

Alleatory		**Markov Decision Process**
Epistemic (Static)		**Reinforcement Learning**
Epistemic (Dynamic)		**POMDP**
Interaction		**Game**

Normal Form Games

- Alice and Bob are working on a homework assignment.
- They can either **share** or **withhold** their knowledge.
- If one player shares knowledge, the other benefits greatly, but the sharer also benefits by getting to test their knowledge

Alice's Payoffs

A \ B	**S**	**W**
S	4	2
W	3	1

Bob's Payoffs

A \ B	**S**	**W**
S	4	3
W	2	1

Alice \ Bob	**S**	**W**
S	4, 4	2, 3
W	3, 2	1, 1

Called a **Normal Form**, **Simple**, or **Bimatrix** Game

Question for today: What **solution concept** should we use for games?

Dominant Strategies

Bob

Alice

	S	**W**
S	4, 4	2, 3
W	3, 2	1, 1

- **Dominant (Pure) Strategy**: Action a is a dominant strategy if it is a best response to every action taken by the other player.
- **Dominant Strategy Equilibrium**: Every player plays a dominant strategy

Definitions

- Action $a^i \in A^i$
- Joint Action $a = (a^1, \ldots, a^k)$
- All Other Actions $a^{-i} = (a^1, \ldots, a^{i-1}, a^{i+1}, \ldots, a^k)$
- Reward $R^i(a)$
- Joint Reward $R(a) = (R^1(a), \ldots, R^k(a))$

Deterministic Best Response:

Action a^i is a deterministic best response to a^{-i} if

$$R^i(a^i, a^{-i}) \geq R^i(a^{i'}, a^{-i}) \quad \forall\, a^{i'}$$

Is the dominant strategy equilibrium always the best outcome for the players?

A more surprising example: The Prisoner's Dilemma

- 2 criminals are captured
- Each can either keep silent (**S**) or testify (**T**)
  - other keeps silent $\rightarrow$ minor conviction (1 year)
  - other testifies $\rightarrow$ major conviction: 4 years
  - testify $\rightarrow$ 1 year removed from sentence

Player 2

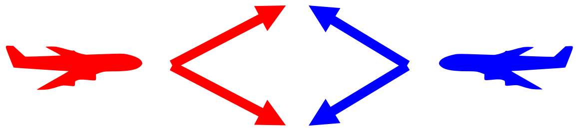
Player 1		**S**	**T**
	S	-1, -1	-4, 0
	T	0, -4	-3, -3

- Dominant strategy for both players is to testify
- Dominant strategy equilibrium is a very bad social result (for the criminals)

Do all simple games have a dominant strategy equilibrium?

Collision Avoidance Game

Example: Airborne Collision Avoidance



Player 2

Player 1

	Up	Down
Up	Collision -5, -5	-1, 0
Down	0, -1	Collision -4, -4

Pure Nash Equilibrium: All players play a deterministic best response.

Which equilibrium is better?

Do all simple games have a pure Nash equilibrium?

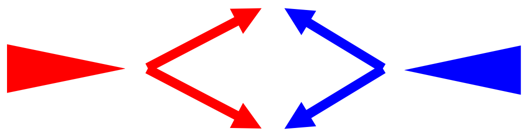
Practice: Find Pure Nash Equilibria

Player 2

Player 1	a	b	c
a	4,4	2,5	0,0
b	5,2	3,3	0,0
c	0,0	0,0	10,10

Missile Defense Game

Missile Defense (simplified)



Defender (columns), **Attacker** (rows)

	Up	Down
Up	Collision -1, 1	1, -1
Down	1, -1	Collision -1, 1

No Pure Nash Equilibrium!

Need a broader solution concept: Mixed Nash equilibrium.

Vocabulary and Notation for Mixed Strategies

	Single Player	Joint
Action	$a^i \in A^i$	$a \in A$
Policy (strategy)	$\pi^i(a^i)$	$\pi(a) = \prod_i \pi^i(a^i)$
Reward	$R^i(a)$	$R(a)$
Utility	$U^i(\pi) = \sum_a R^i(a)\pi(a)$	$U(\pi) = \sum_a R(a)\pi(a)$

Two Player Zero Sum:

$$R^1(a) + R^2(a) = 0 \quad \forall a$$

Cooperative:

$$R^i(a) = R^j(a) \quad \forall\, i, j, a$$

("General sum" used to indicate that the game is not limited to zero-sum or coop.)

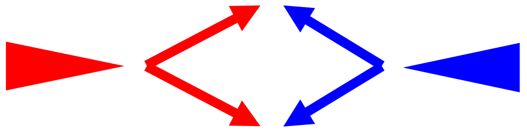
Best Response: Given a joint policy of all other agents, π^{-i}, a best response is a policy π^i that satisfies

$$U^i\left(\pi^i, \pi^{-i}\right) \geq U^i\left(\pi^{i'}, \pi^{-i}\right)$$

for all other $\pi^{i'}$.

Missile Defense Game

Missile Defense (simplified)



Defender

Attacker

		Up	Down
Attacker	Up	Collision -1, 1	1, -1
	Down	1, -1	Collision -1, 1

- A ***Nash equilibrium*** is a joint policy in which **all agents play a best response**

Rock-paper scissors

1. Guess the Nash Equilibrium argument
2. Make a qualitative argument that this is an NE based on best responses

agent 1 \ agent 2	rock	paper	scissors
rock	$0, 0$	$-1, 1$	$1, -1$
paper	$1, -1$	$0, 0$	$-1, 1$
scissors	$-1, 1$	$1, -1$	$0, 0$

Do all simple games have at least one Nash equilibrium? Yes!! (might be mixed)

Every finite game has a Nash Equilibrium

Kakutani's fixed-point theorem

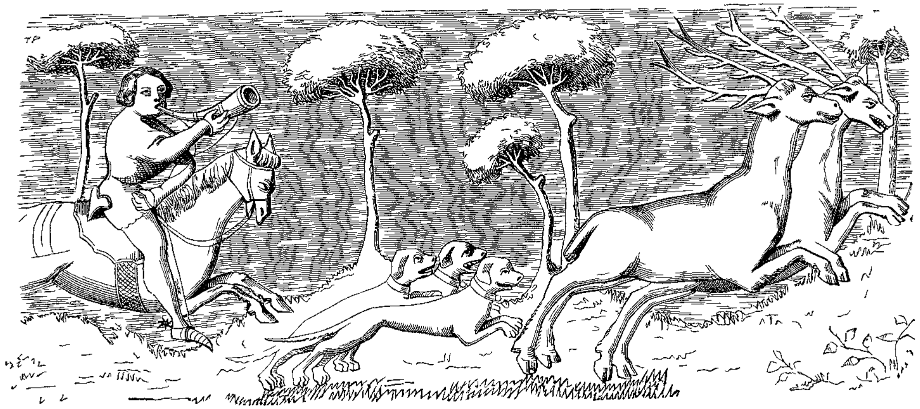
A correspondence f: X → X has a fixed point (i.e., $\mathbf{x} \in f(\mathbf{x})$ for some $\mathbf{x} \in X$) if all of the following conditions hold.

(1) X is a non-empty, closed, bounded, and convex set.
(2) $f(\mathbf{x})$ is non-empty for any $\mathbf{x}$.
(3) $f(\mathbf{x})$ is convex for any $\mathbf{x}$.
(4) The set $\{ (\mathbf{x}, \mathbf{y}) \mid \mathbf{y} \in f(\mathbf{x}) \}$ is closed.

- Let x be a joint strategy, π.
- Let f be BR, that is, the best response operator
- A fixed point of BR is a Nash Equilibrium
- The BR operator and policy space for finite games meet the conditions above
- BR has a fixed point for every finite game, i.e. every finite game has a Nash Equilibrium

Not on Exam

Calculating Mixed Nash



	Stag	Hare
Stag	4, 4	1, 3
Hare	3, 1	2, 2

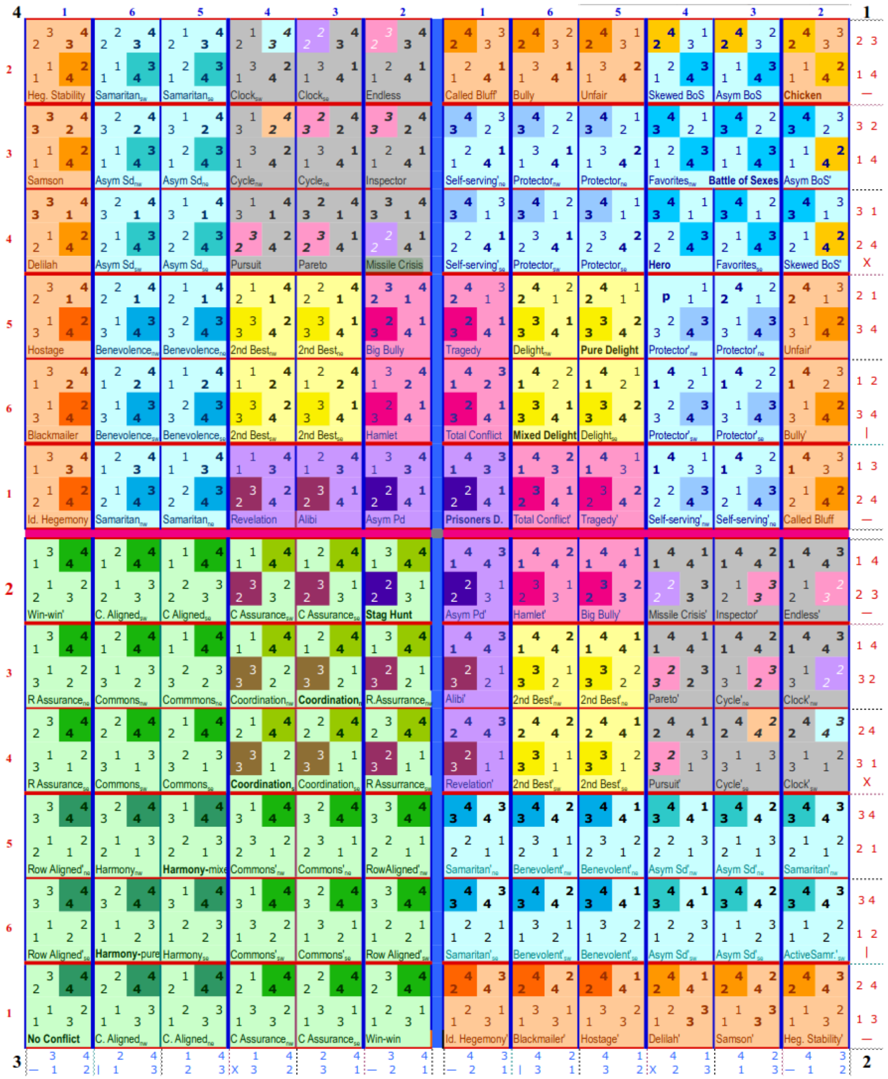
- In a Mixed Nash Equilibrium, players must be *indifferent* between two or more actions
- (In large games, finding the support of the mixed strategies is the hard part)

General approach to find Nash Equilibria

$$
\begin{aligned}
\underset{\pi, U}{\text{minimize}} \quad & \sum_i \left(U^i - U^i(\pi) \right) \\
\text{subject to} \quad & U^i \geq U^i(a^i, \pi^{-i}) \text{ for all } i, a^i \\
& \sum_{a^i} \pi^i(a^i) = 1 \text{ for all } i \\
& \pi^i(a^i) \geq 0 \text{ for all } i, a^i
\end{aligned}
$$

Not on Exam

Topology of bimatrix games:



Algorithms that use best response

Iterated Best Response: randomly cycle between agents who play the best response for the current policy (converges to Nash for certain narrow classes of games)

Fictitious Play:

1. Estimate maximum likelihood policies for opponents:
$$\pi^j(a^j) \propto N(j, a^j)$$
2. Play best response to estimated policy

(converges to Nash for wider class of games, notably zero-sum)

Bach or Stravinsky

- Two people want to go to a concert
- P1 prefers Bach, P2 Stravinsky

	B	S
B	2, 1	0,0
S	0, 0	1, 2

Correlated Equilibrium

- A *correlated joint policy* is a single distribution over the joint actions of all agents.
- A *correlated equilibrium* is a correlated joint policy where no agent *i* can increase their expected utility by deviating from the current policy.
- Easier to find than Nash equilibrium (Linear Program)

https://youtube.com/shorts/w3q77ZZIqwA
?si=J8H6L6W5kTRs-mUx

Recap

- Games provide a mathematical framework for analyzing interaction between rational agents
- Games may not have a single "optimal" solution; instead there are equilibria that may not be rankable
- If every player is playing a best response, that joint policy is a Nash Equilibrium
- Every finite game has at least one Nash Equilibrium (pure or mixed)
- Mixed Nash equillibria occur when players are indifferent between two outcomes